

PERCEPTRÓN SIMPLE Y MULTICAPA

Sistemas de Inteligencia Artificial – TP 3

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EXERCISE 1

Develop a lightweight model called **TinyModel** for CompanyX using Knowledge Distillation to match the performance of the current model they are using at a lower cost. Implement and evaluate both linear and non-linear simple perceptrons to analyze learning behaviors, underfitting, and capacity saturation, followed by a full generalization study and detection threshold recommendation.

EXERCISE 2

Implement a multilayer perceptron (MLP) to solve the task of automatically classifying handwritten digits (0–9). Use the provided training dataset to fine-tune parameters and hyperparameters, comparing different learning rates, network architectures, and optimization methods, while simultaneously evaluating the model's generalization performance using a separate test dataset.

EXERCISE 3

Improve your classification process to meet CompanyX's stricter accuracy target of at least 98% by leveraging the new, expanded dataset. Document the advanced methods, changes, and external factors that contributed to the performance improvement compared to the previous version.

Q1:

Q2:

Q3: ADEMÁS DE SUS PROPIAS TÉCNICAS, ¿EXISTEN OTROS FACTORES QUE INFLUYERON EN EL CAMBIO DE RENDIMIENTO ENTRE ESTE EJERCICIO Y EL ANTERIOR?

The Data:

- Both datasets:

- 28 × 28 pixel images → 784 input features
- Pixel values already scaled between 0 and 1
- Very similar overall pixel distributions

Key difference: Class distribution

- digits.csv
 - No samples of digit 8
 - Only 271 samples of digit 5
 - Most other classes: ~1,450–1,700 samples
- more_digits.csv
 - Contains all 10 digit classes
 - 585 samples of digit 8
 - 542 samples of digit 5
- Effect on performance: Exercise 2 model has never seen digit 8 during training, and digit 5 is also underrepresented
 - more_digits.csv provides: More training samples, better class coverage and training examples for the previously unseen digit 8
 - Performance improvements may therefore come from better data, not only from changes to the model or training techniques.